

POST-PROCESSING

Post-processing

- Visualization
 - The **human eye** is a powerful analytical tool
 - If we visualize the data properly, we can discover patterns and demonstrate trends
 - Visualization is the way to present the data so that patterns can be seen
 - E.g., histograms and plots are a form of visualization
 - There are multiple techniques (a field on its own)

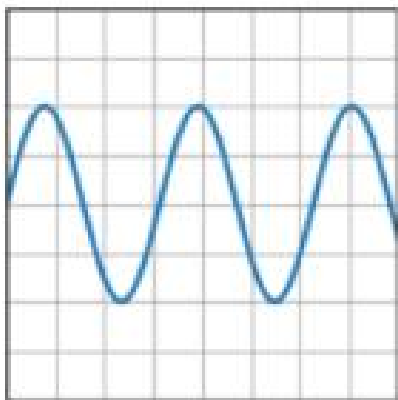
Visualization on a map

- John Snow, London 1854



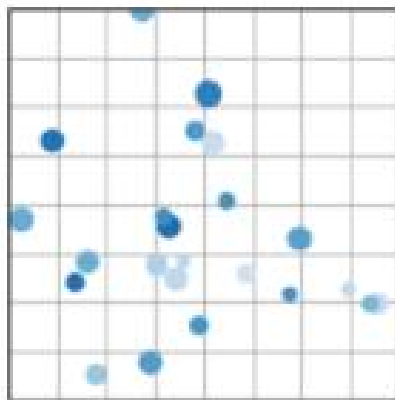
Basic

Basic plot types, usually y versus x.



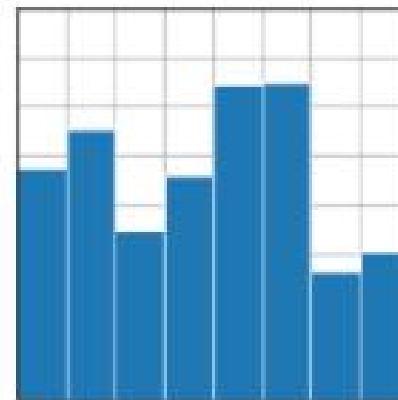
`plot(x, y)`

折线图



`scatter(x, y)`

散点图

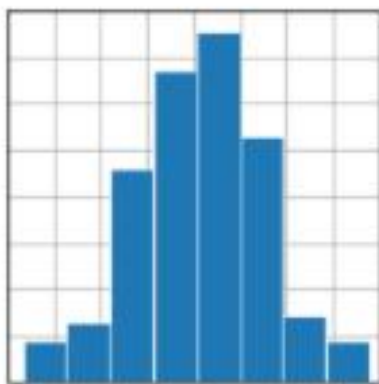


`bar(x, height) / barh(y, width)`

条形图

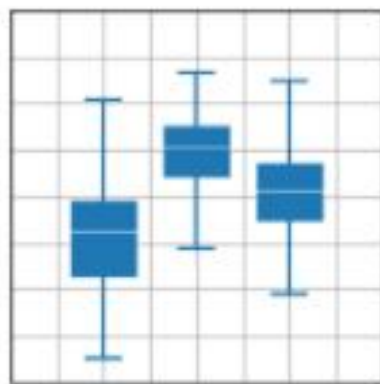
Statistics plots

Plots for statistical analysis. <https://matplotlib.org/>



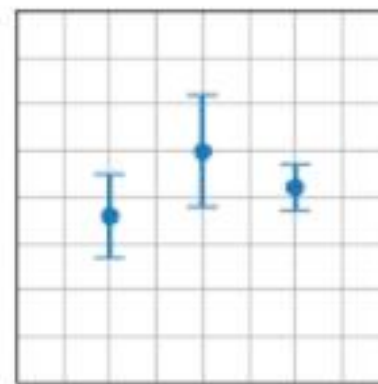
`hist(x)`

Histogram 直方图
分布展示



`boxplot(X)`

箱形图: max,min,median, 25th percentile, 75th percentile
误差线: 表示标准差或标准误



`errorbar(x, y, yerr, xerr)`



`pie(x)`

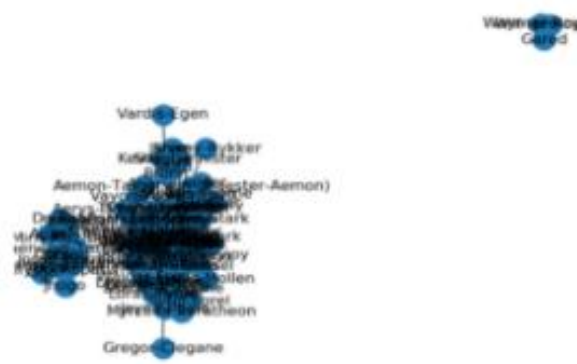
饼图

标准误 Standard Error: 衡量样本均值与总体均值之间的差异, 样本的标准差除以样本容量的平方根

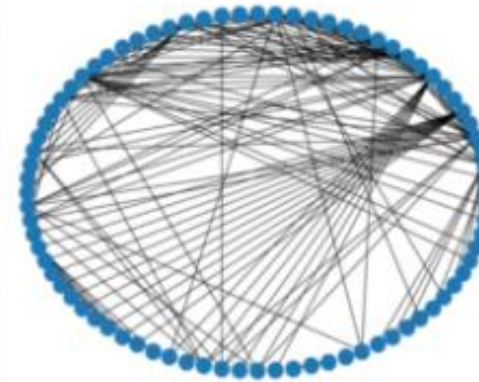
`nx.draw(G)`



`nx.draw_networkx(G)`



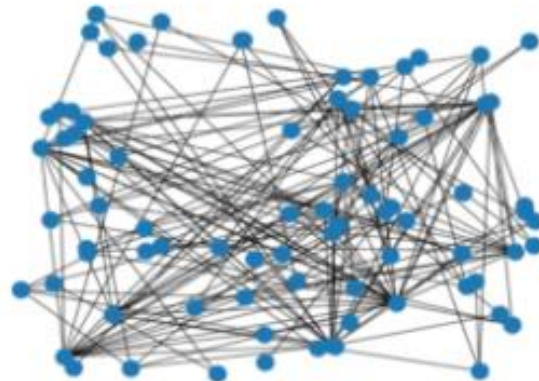
`nx.draw_circular(G)`



`nx.draw_kamada_kawai(G)`



`nx.draw_random(G)`



`nx.draw_spring(G)`



Dimensionality Reduction

- The human eye is limited to processing visualizations in two (at most three) dimensions
- One of the great challenges in visualization is to visualize **high-dimensional data** into a **two-dimensional** space
 - Dimensionality reduction
 - Distance preserving embeddings
- Dimensionality reduction is also a **preprocessing** technique:
 - Reduce the amount of data
 - Extract the useful information.

Example

- Consider the following 6-dimensional dataset

$$D = \begin{bmatrix} 1 & 2 & 3 & 0 & 0 & 0 \\ 2 & 4 & 6 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 2 & 4 & 6 \\ 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 4 & 6 & 2 & 4 & 6 \end{bmatrix}$$

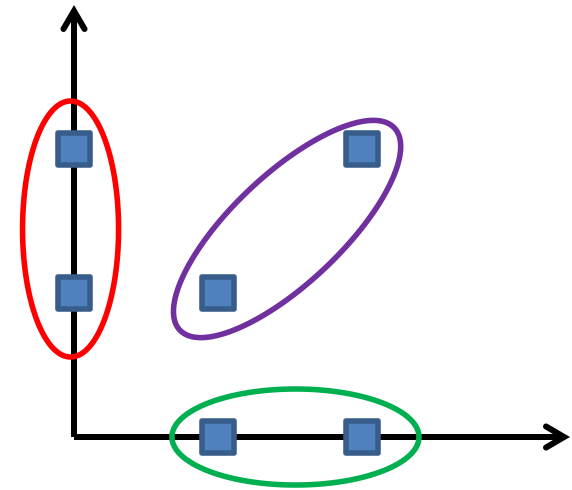
- What do you **observe**? Can we reduce the dimension of the data?

Example

- Each row is a **multiple** of two **vectors**
 - $x = [1, 2, 3, 0, 0, 0]$
 - $y = [0, 0, 0, 1, 2, 3]$
- We can rewrite D as

$$D = \begin{array}{cc} & \begin{matrix} x & y \end{matrix} \\ \begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \\ 1 \\ 2 \end{bmatrix} & \begin{bmatrix} 0 \\ 0 \\ 1 \\ 2 \\ 1 \\ 2 \end{bmatrix} \end{array}$$

$$D = \begin{bmatrix} 1 & 2 & 3 & 0 & 0 & 0 \\ 2 & 4 & 6 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 2 & 4 & 6 \\ 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 4 & 6 & 2 & 4 & 6 \end{bmatrix}$$



Three types of data points

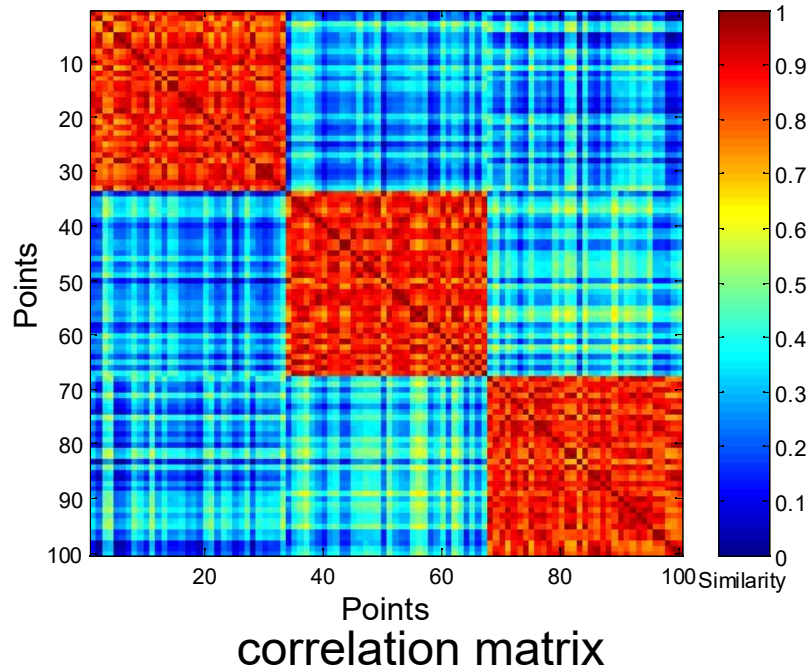
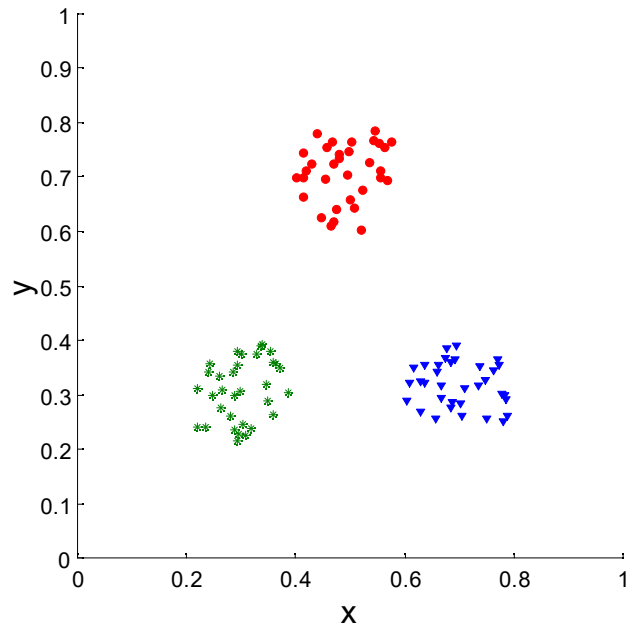
Word Clouds

- A fancy way to visualize a document or collection of documents.



Heatmaps

- Plot a point-to-point similarity matrix using a heatmap:
 - Deep red = high values (hot)
 - Dark blue = low values (cold)

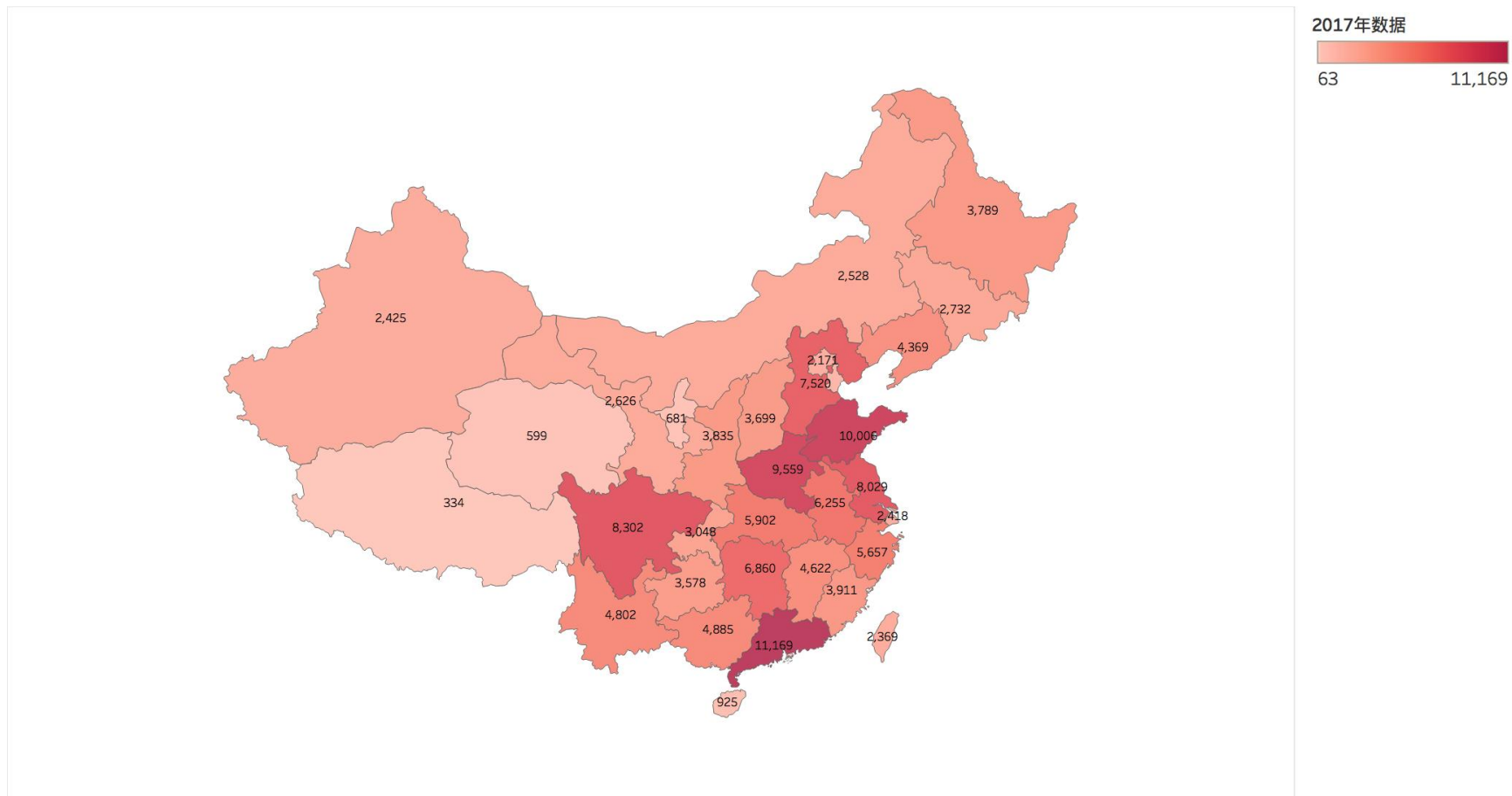


The clustering structure becomes clear in the heatmap

Heatmaps

A very popular way to visualize data

2017各省人口数据热力图



基于 经度(生成) 和 纬度(生成) 的地图。 颜色显示 2017年数据 总和。 为 地区 显示了详细信息。

Statistical Significance

- When we extract knowledge from a large dataset we need to make sure that what we found is not an **artifact of randomness**
 - E.g., we find that many people buy milk and beer together.
 - But many (more) people buy milk and beer **independently**
- Statistical tests compare the results of an experiment with those generated by a **null hypothesis**
 - E.g., a null hypothesis is that people select items independently.
- A result is interesting if it cannot be produced by **randomness**.
 - An important problem is to define the null hypothesis correctly: What is random?

P-value

- A p-value measures the **probability of obtaining the observed results, assuming that the null hypothesis is true**. The lower the p-value, the greater the statistical significance of the observed difference.
- 例子：连续抛一枚硬币5次，每次都正面朝上，判断硬币是否均匀。
 - **null hypothesis**：硬币是均匀的（正面朝上和反面朝上的概率一样，各50%）
 - 如果原假设成立，结果（5次都是正面朝上）发生的概率是 $0.5^5=0.03125$. 所以 $p\text{-value} = 0.03125$ ，可以拒绝原假设。

EXPLORATORY DATA ANALYSIS

What does my data look like?

Exploratory analysis of data

- **Summary statistics**: numbers that summarize properties of the data
- Summarized properties include **frequency** (频率), **location** (定位) and **spread** (离散程度)
 - Examples: location - mean
spread - standard deviation
- Most summary statistics can be calculated in a single pass through the data
- Computing **data statistics** is one of the first steps in understanding our data

Frequency and Mode

- The **frequency** of an attribute value is the percentage of time the value occurs in the data set
 - For example, given the attribute 'gender' and a representative population of people, the gender 'female' occurs about 50% of the time.
- The **mode** (众数) of an attribute is the most frequent attribute value
- The notions of frequency and mode are typically used with categorical data
- We can visualize the data frequencies using a **value histogram**

Example

<i>Tid</i>	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	10000K	Yes
6	No	NULL	60K	No
7	Yes	Divorced	220K	NULL
8	No	Single	85K	Yes
9	No	Married	90K	No
10	No	Single	90K	No

Marital Status

Single	Married	Divorced	NULL
4	3	2	1

Mode: Single

Example

<i>Tid</i>	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
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3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	10000K	Yes
6	No	NULL	60K	No
7	Yes	Divorced	220K	NULL
8	No	Single	85K	Yes
9	No	Married	90K	No
10	No	Single	90K	No

Marital Status

Single	Married	Divorced	NULL
40%	30%	20%	10%

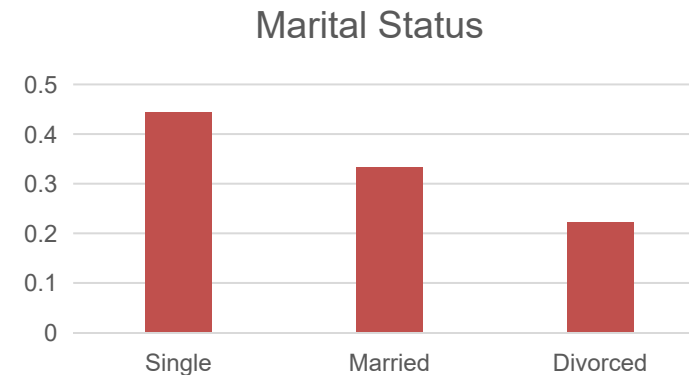
Example

<i>Tid</i>	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	10000K	Yes
6	No	NULL	60K	No
7	Yes	Divorced	220K	NULL
8	No	Single	85K	Yes
9	No	Married	90K	No
10	No	Single	90K	No

We can choose to ignore NULL values

Marital Status

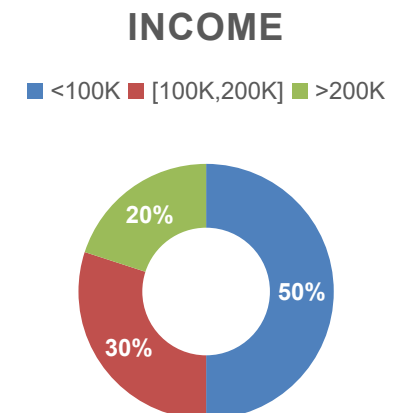
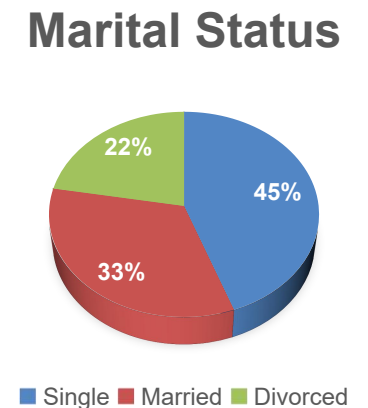
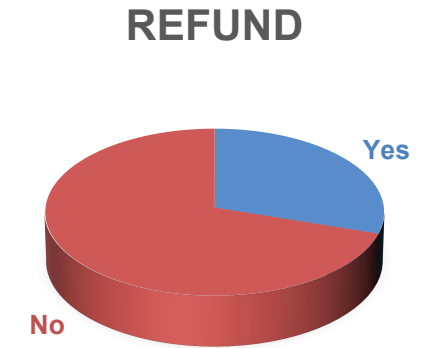
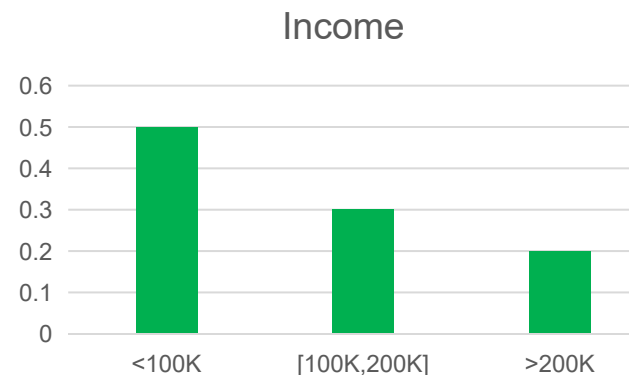
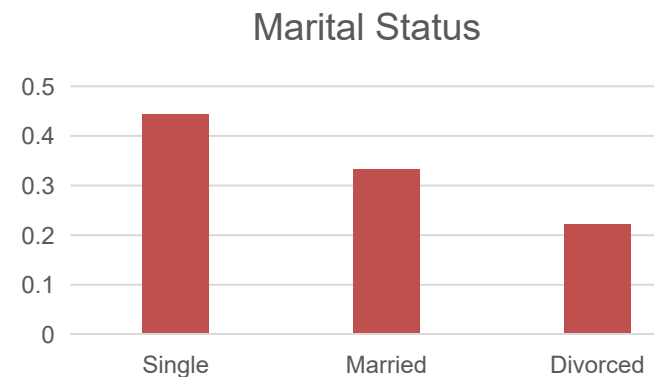
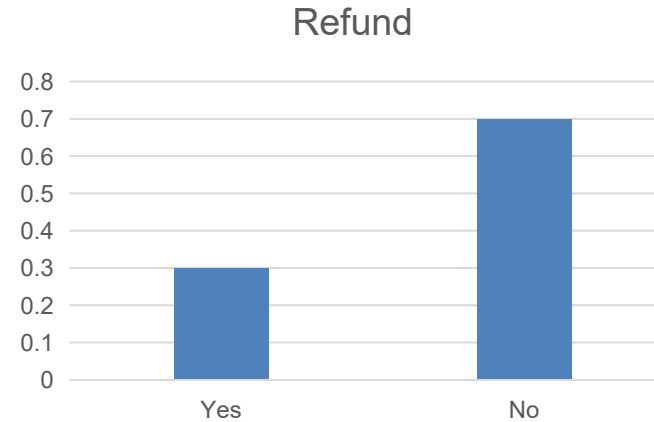
Single	Married	Divorced
44%	33%	22%



Data histograms

<i>Tid</i>	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	10000K	Yes
6	No	NULL	60K	No
7	Yes	Divorced	220K	NULL
8	No	Single	85K	Yes
9	No	Married	90K	No
10	No	Single	90K	No

Use **binning** for numerical values



Percentiles

- For continuous data, the notion of a percentile (百分位数) is more useful.

Given an ordinal or continuous attribute x and a number p between 0 and 100, the p^{th} percentile (第 p 百分位数) is a value x_p of x such that $p\%$ of the observed values of x are less than or equal to x_p .

- For instance, the 80th percentile is the value $x_{80\%}$ that is greater or equal than 80% of all the values of x we have in our data.

Example

<i>Tid</i>	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
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6	No	NULL	60K	No
7	Yes	Divorced	220K	NULL
8	No	Single	85K	Yes
9	No	Married	90K	No
10	No	Single	90K	No

Taxable Income
10000K
220K
125K
120K
100K
90K
90K
85K
70K
60K

$$x_{80\%} = 125K$$

Measures of Location: Mean and Median

- The **mean** is the most common measure of the location of a set of points.

$$\text{mean}(x) = \bar{x} = \frac{1}{m} \sum_{i=1}^m x_i$$

- However, the mean is very sensitive to outliers.
- Thus, the **median** is also commonly used.

$$\text{median}(x) = \begin{cases} x_{(r+1)} & \text{if } m \text{ is odd, i.e., } m = 2r + 1 \\ \frac{1}{2}(x_{(r)} + x_{(r+1)}) & \text{if } m \text{ is even, i.e., } m = 2r \end{cases}$$

- Or the **trimmed mean** (裁剪平均值) : the mean after removing min and max values

Example

<i>Tid</i>	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	10000K	Yes
6	No	NULL	60K	No
7	Yes	Divorced	220K	NULL
8	No	Single	85K	Yes
9	No	Married	90K	No
10	No	Single	90K	No

Mean: 1090K

Trimmed mean (remove min, max): 105K

Median: $(90+100)/2 = 95K$

Measures of Spread: Range and Variance

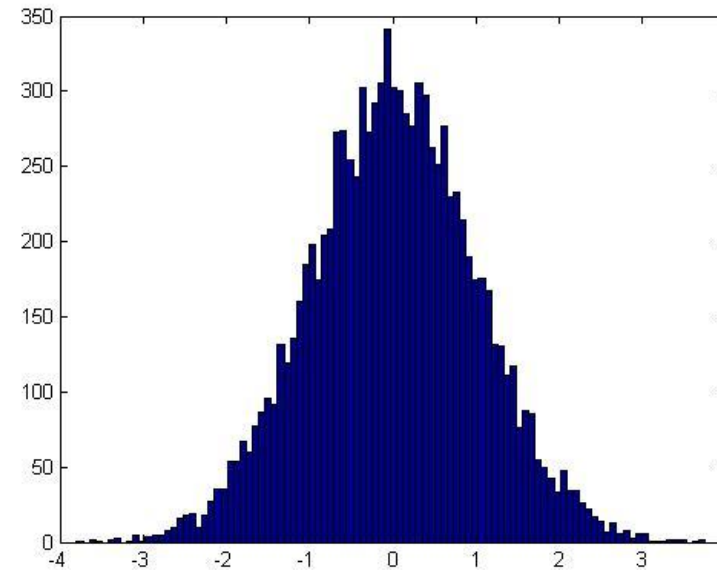
- **Range** (极差、全距) is the difference between the **max** and **min**
- The **variance** (方差) or **standard deviation** (标准差) is the most common measure of the spread of a set of points.

$$var(x) = \frac{1}{m} \sum_{i=1}^m (x - \bar{x})^2$$

$$\sigma(x) = \sqrt{var(x)}$$

Normal Distribution

- $$\phi(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

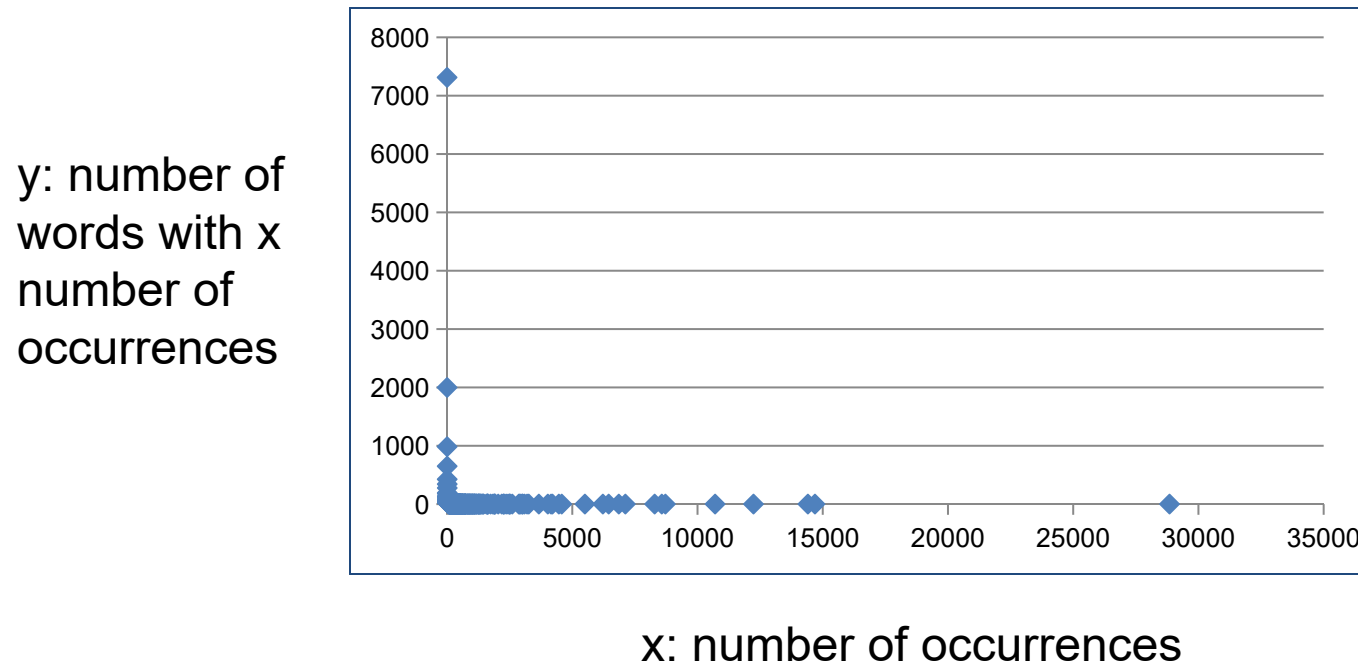


This is a value histogram

- An important distribution that characterizes many quantities and has a central role in probabilities and statistics.
- Fully characterized by the mean μ and standard deviation σ

Not everything is normally distributed

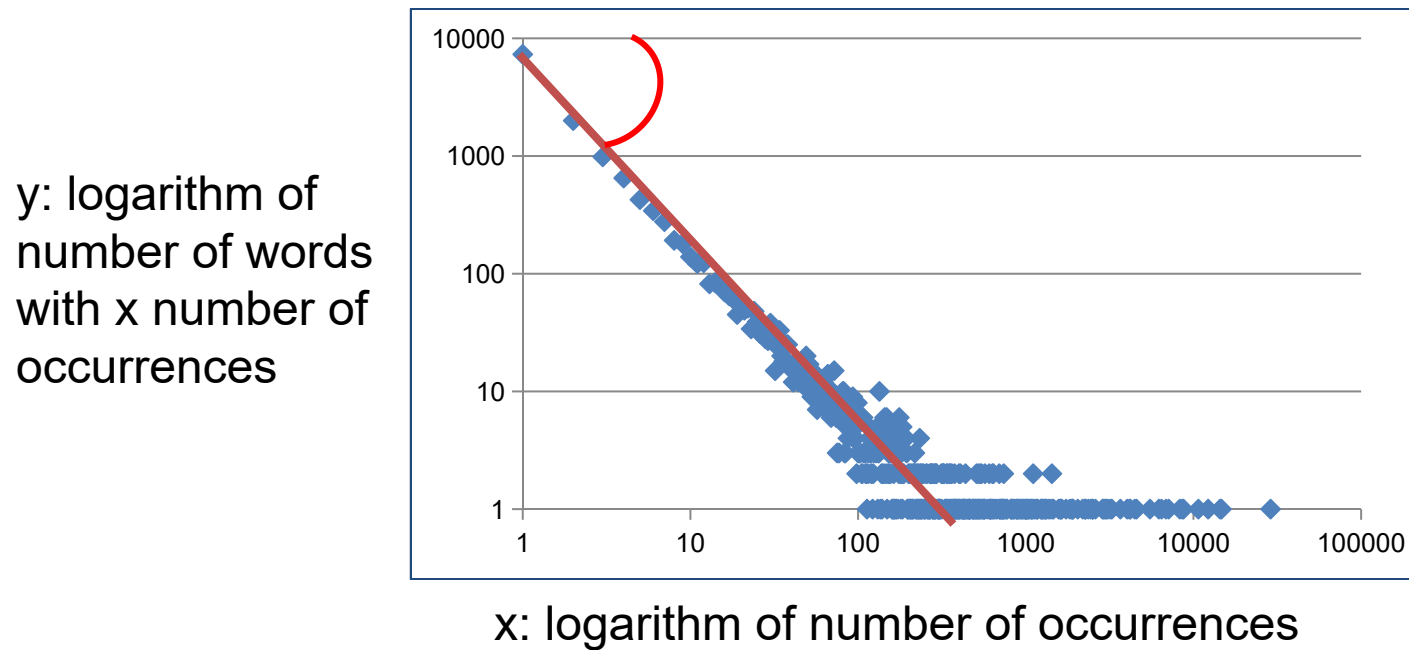
- Plot of number of words with x number of occurrences



- If this was a normal distribution we would not have number of occurrences as large as **28K**

Power-law distribution

- We can understand the distribution of words if we take the **log-log** plot



Linear relationship in the log-log space

$$\log p(x = k) = -\alpha \log k$$

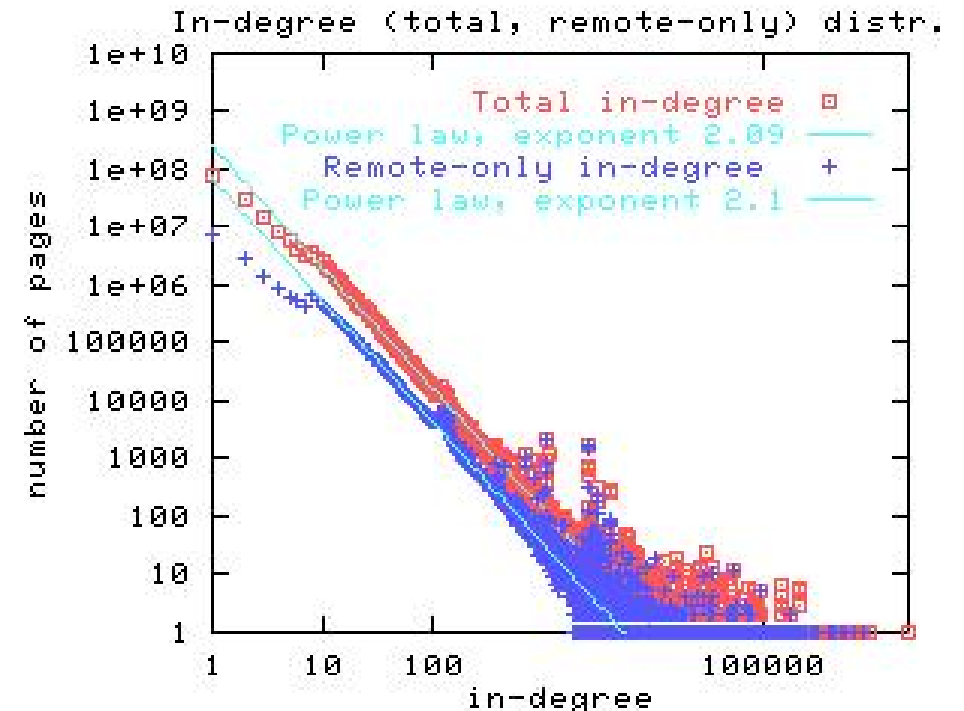
Power-law distribution:

$$p(k) = k^{-\alpha}$$

The **slope** of the line gives us the exponent **α**

Power-laws are everywhere

- number of friends in social networks, number of occurrences of words, city sizes, income distribution, popularity of products and movies
 - Signature of human activity?
 - A mechanism that explains everything?
 - Rich get richer process



Attribute relationships

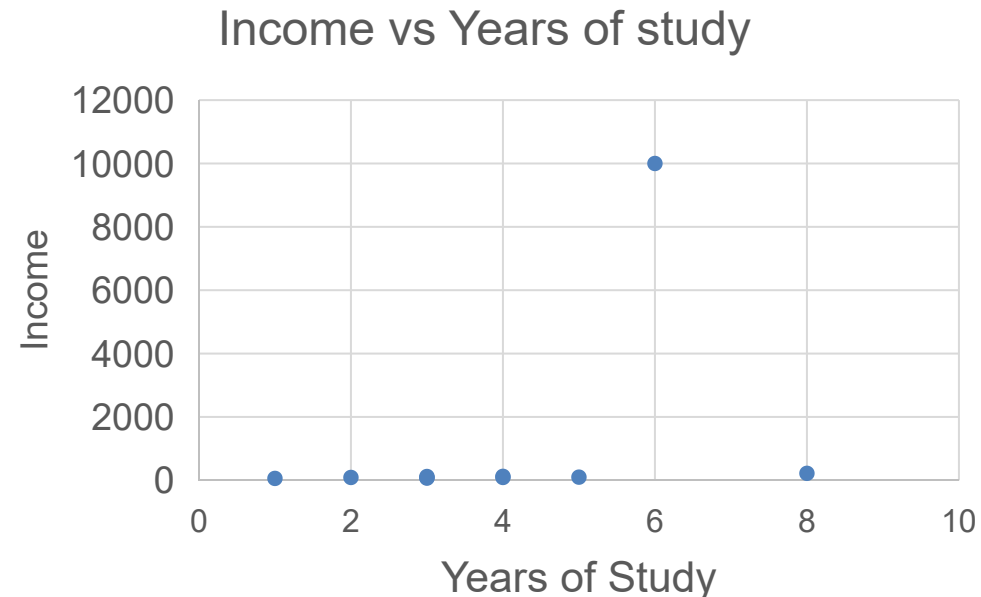
- In many cases it is interesting to look at two attributes together to understand if they are correlated
 - Is there a relationship between years of study and income?
- How do we visualize these relationships?

Correlating numerical attributes

<i>Tid</i>	Refund	Marital Status	Taxable Income	Years of Study
1	Yes	Single	125K	4
2	No	Married	100K	5
3	No	Single	70K	3
4	Yes	Married	120K	3
5	No	Divorced	10000K	6
6	No	NULL	60K	1
7	Yes	Divorced	220K	8
8	No	Single	85K	3
9	No	Married	90K	2
10	No	Single	90K	4

Scatter plot:

X axis is one attribute, Y axis is the other
For each entry we have two values
Plot the entries as two-dimensional points



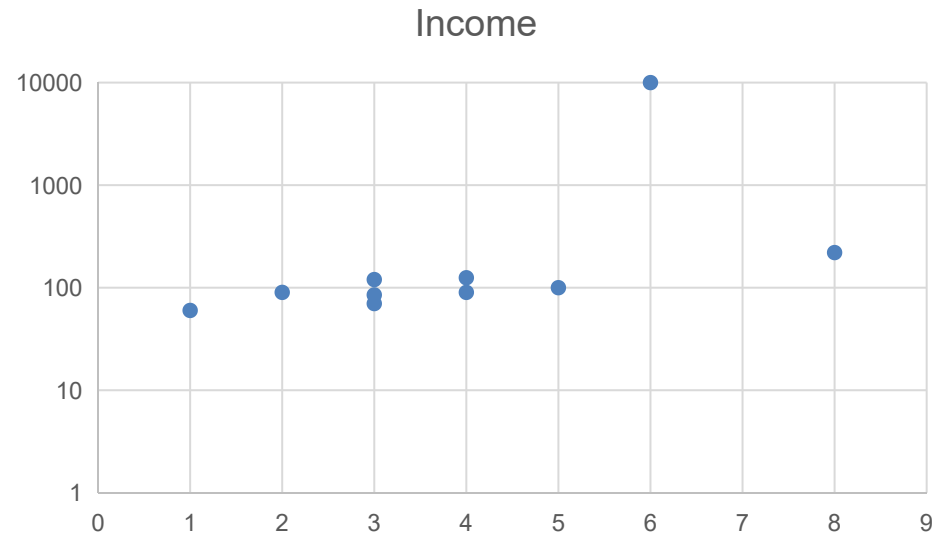
Correlating numerical attributes

<i>Tid</i>	Refund	Marital Status	Taxable Income	Years of Study
1	Yes	Single	125K	4
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3	No	Single	70K	3
4	Yes	Married	120K	3
5	No	Divorced	10000K	6
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10	No	Single	90K	4

Scatter plot:

X axis is one attribute, Y axis is the other
For each entry we have two values
Plot the entries as two-dimensional points

Log-scale in y-axis makes the plot look a little better



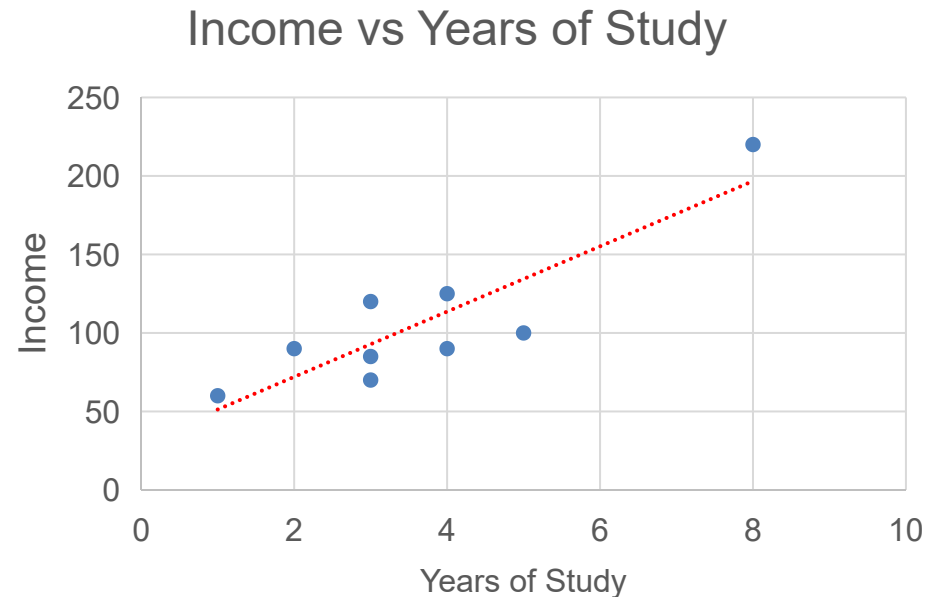
Plotting attributes against each other

Tid	Refund	Marital Status	Taxable Income	Years of Study
1	Yes	Single	125K	4
2	No	Married	100K	5
3	No	Single	70K	3
4	Yes	Married	120K	3
5	No	Divorced	10000K	6
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Scatter plot:

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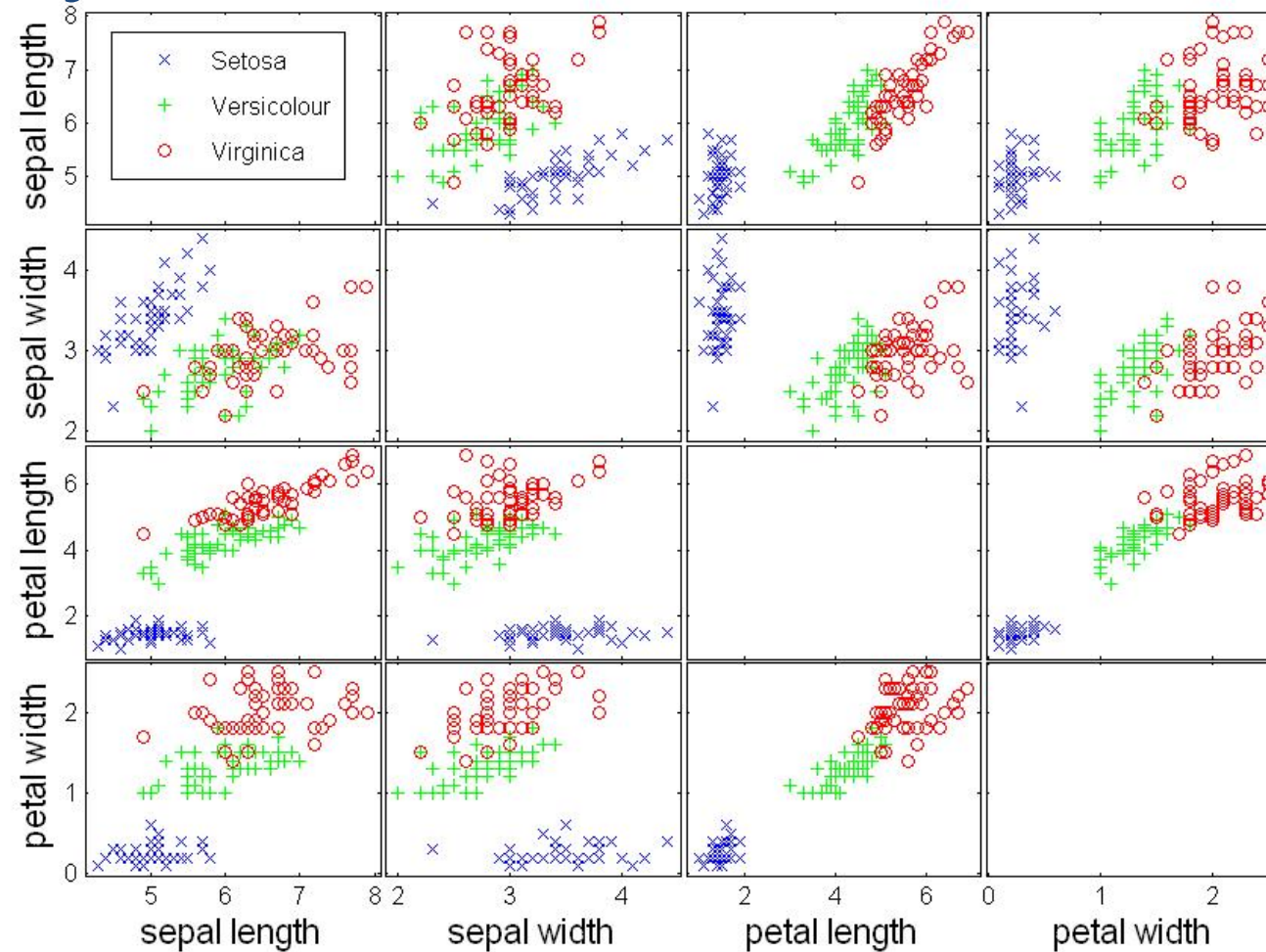
After removing the outlier value there is a clear correlation



Scatter Plot Array of Iris Attributes

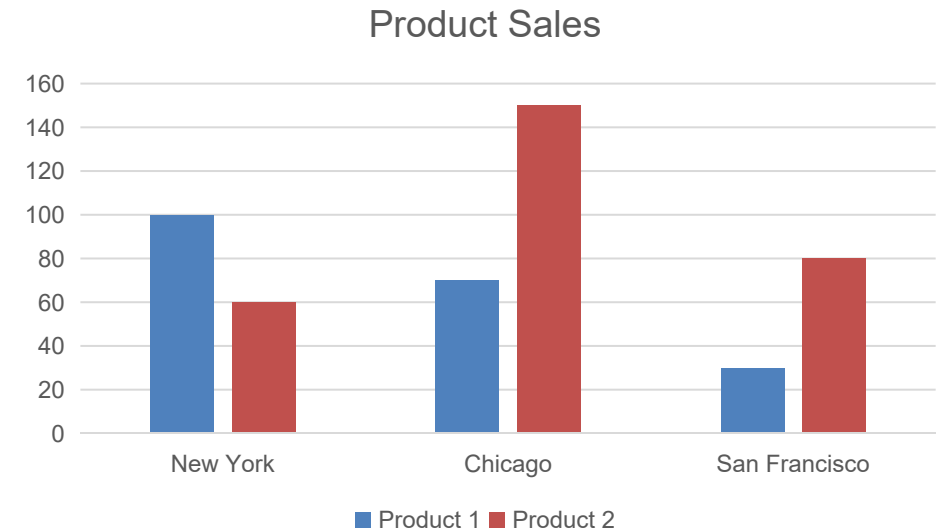
For multiple attribute pairs

from sklearn import datasets
iris = datasets.load_iris()



Plotting attributes together

City	Product 1	Product 2
New York	100	60
Chicago	70	150
San Francisco	30	80

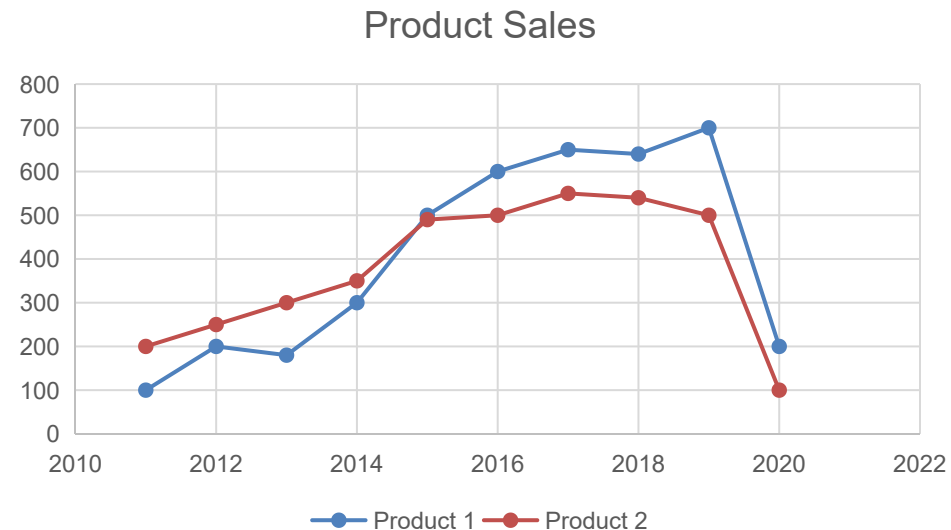


How would you visualize the differences between the product sales per city?

Plotting attributes together

Year	Product 1	Product 2
2011	100	200
2012	200	250
2013	180	300
2014	300	350
2015	500	490
2016	600	500
2017	650	550
2018	640	540
2019	700	500
2020	200	100

How would you visualize the differences between the product sales over time?



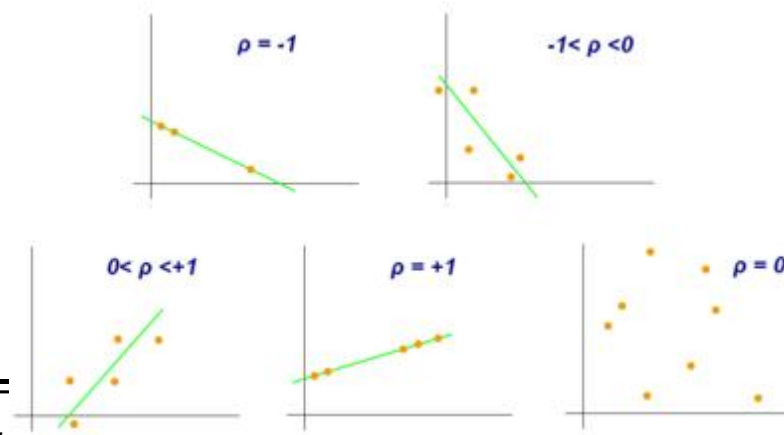
Measuring correlation

- **Pearson correlation coefficient**: measures the extent to which two variables are **linearly correlated**

- $X = \{x_1, \dots, x_n\}$
 - $Y = \{y_1, \dots, y_n\}$
- Must have **pairs** of observations

$$\text{corr}(X, Y) = \frac{\frac{1}{n} \sum_i (x_i - \mu_X)(y_i - \mu_Y)}{\sqrt{\frac{\sum_i (x_i - \mu_X)^2}{n}} \sqrt{\frac{\sum_i (y_i - \mu_Y)^2}{n}}} = \frac{\sum_i (x_i - \mu_X)(y_i - \mu_Y)}{\sqrt{\sum_i (x_i - \mu_X)^2} \sqrt{\sum_i (y_i - \mu_Y)^2}}$$

- It comes with a **p-value**
 - The p-value is the probability that the correlation was by chance.
- Assumes no outliers and that the variables are normally distributed



- **Spearman rank correlation coefficient:** tells us if two variable are rank-correlated
 - They place items in the same order – Pearson correlation of the rank vectors
 - For ranking without ties it looks at the differences between the ranks of the same items

The scores for nine students in physics and math are as follows:

Physics: 35, 23, 47, 17, 10, 43, 9, 6, 28

Mathematics: 30, 33, 45, 23, 8, 49, 12, 4, 31

Physics	Rank	Math	Rank
35	3	30	5
23	5	33	3
47	1	45	2
17	6	23	6
10	7	8	8
43	2	49	1
9	8	12	7
6	9	4	9
28	4	31	4

	姓名	项目标题	Paper Presentation论文选题
1	申士愷 李昀峰 郑证	中文姓名性别预测	For the Underrepresented in Gender Bias Research: Chinese Name Gender Prediction with Heterogeneous Graph Attention Network
2	高烨 虞果 张博涵 李彧轩	基于新闻的金融市场预测	Textual analysis of stock market prediction using breaking financial news: The AZFin text system
3	马蒙麒 徐通懿 孙文妍	基于新闻的金融市场预测	Textual analysis of stock market prediction using breaking financial news: The AZFin text system
4	尹椽 孙博飞 万宇昂 陈英恺 吴沛达	全球照片地理分布挖掘	Mapping the world's photos
5	钱璿 熊章智 李明瑞	基于新闻的金融市场预测	Mining the web to predict future events
6	刘蓁 李沐阳 李元瑄 刘嘉芙 吕承轩		



分类	主题	标题
多智能体系统	多智能体股票交易	HedgeAgents: A Balanced-aware Multi-agent Financial Trading System
多智能体系统	金融市场行为模拟	TwinMarket: A Scalable Behavioral and Social Simulation for Financial Markets
多智能体系统	自主式辩论系统	An autonomous debating system Nature
多智能体系统	生成式智能体模拟人类行为	Generative Agents: Interactive Simulacra of Human Behavior
多智能体系统	数据科学自动化智能体	DatawiseAgent: A Notebook-Centric LLM Agent Framework for Adaptive and Robust Data Science Automation
股票预测	基于新闻的金融市场预测	Textual analysis of stock market prediction using breaking financial news: The AZFin text system
公司分析	初创企业成功预测	Help Me Screen: Analyzing and Predicting the Success of Start-ups in Dynamic Venture Capital Networks
事件预测	基于网络数据的未来事件预测	Mining the web to predict future events
人口流动预测	基于LinkedIn的国际迁移意愿分析	Openness to Migrate Internationally for a Job: Evidence from LinkedIn Data in Europe
人口流动预测	全球劳动力流动网络分析	Global labor flow network reveals the hierarchical organization and dynamics of geo-industrial clusters Nature Communications
人口流动预测	人类移动性的普适规律	The universal visitation law of human mobility
多模态	古诗词多模态情感分析	Picturized and Recited with Dialects: A Multimodal Chinese Representation Framework for Sentiment Analysis of Classical Chinese Poetry
跨学科/政治学	社交媒体与政治传播分析	Why does the president tweet this? Discovering reasons and contexts for politicians’ tweets from news articles
跨学科/政治学	利用大数据改进选举预测	Improving election prediction internationally
跨学科/计算社会学	基于百科的人生轨迹挖掘	Paths of A Million People: Extracting Life Trajectories from Wikipedia
跨学科/公平性研究	中文姓名性别预测	For the Underrepresented in Gender Bias Research: Chinese Name Gender Prediction with Heterogeneous Graph Attention Network
跨学科/文化地理学	地域菜系相似性分析	Geography and Similarity of Regional Cuisines in China
跨学科/经济复杂性	产品空间与国家发展	The product space conditions the development of nations
跨学科/多媒体分析	全球照片地理分布挖掘	Mapping the world’s photos
跨学科/流行病学	搜索引擎数据在疫情监测中的应用	两篇一起Detecting influenza epidemics using search engine query data 与The Parable of Google Flu: Traps in Big Data Analysis
跨学科/人机交互	维基百科用户浏览行为分析	两篇一起A Large-Scale Characterization of How Readers Browse Wikipedia 与 On the Evolution of Wikipedia

- Presentation内容通常应包括：论文背景、相关文献、问题介绍、方法 motivation（对于方法创新的文章）、方法介绍、实验与实验结果（或发现、结论）、自己的想法（启发、如何改进，等等）。
- 如果是较早的论文，请思考现在如果做类似的工作，应该如何入手。
- 如果是综合类期刊论文，也请阅读对于理解正文所必要的补充材料内容。
- 也可自选论文，原则上来源应为CCF A类B类期刊/会议（如WWW, KDD, WSDM, CIKM, ICWSM, AAI, IJCAI, ICDM）的长文，数据挖掘相关。
- 请于**3月26日（第4周周四）晚23:59前**登记论文选题 <https://docs.qq.com/sheet/DWUxCVG9YRnFpTUpL?tab=BB08J2>，如自选论文，presentation前，也请至少提前三周（3月26日，第4周周四前）发给老师审核确认。
- 请于**4月16日（第7周周四）晚23:59前**将PPT提交至BB。（PPT允许现场更新）
- Presentation时间为**4月20日（第8周周一）早10:15-11:55**
- **英文PPT**，每组10分钟以内presentation, 3分钟以内QA。